

Q1: 24 Feb (Shift 1) - Single Correct

Given below are two statements :

Statement-I: Two photons having equal linear momenta have equal wavelengths.

Statement-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Statement-I is false but Statement-II is true
- (2) Both Statement-I and Statement-II are true
- (3) Both Statement-I and Statement-II are false
- (4) Statement-I is true but Statement-II is false

Q2: 24 Feb (Shift 1) - Numerical

A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_{p} denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{\text{Li}}}{\lambda_{\text{p}}}$ is $x \times 10^{-1}$. The value of x is ____.

[Rounded off to the nearest integer] [Mass of $\text{Li}^{3+} = 8.3$ mass of proton]

Q3: 24 Feb (Shift 2) - Single Correct

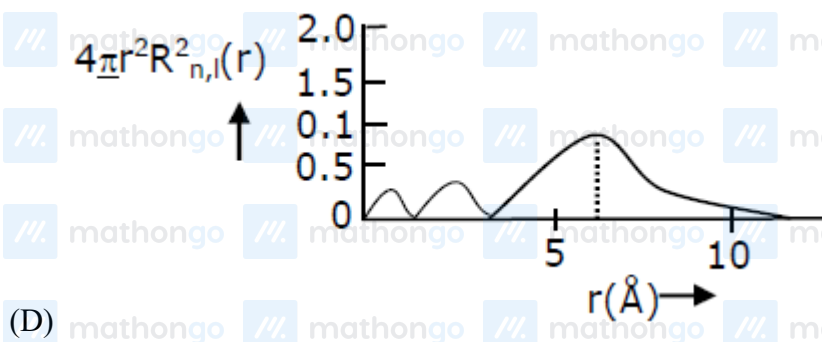
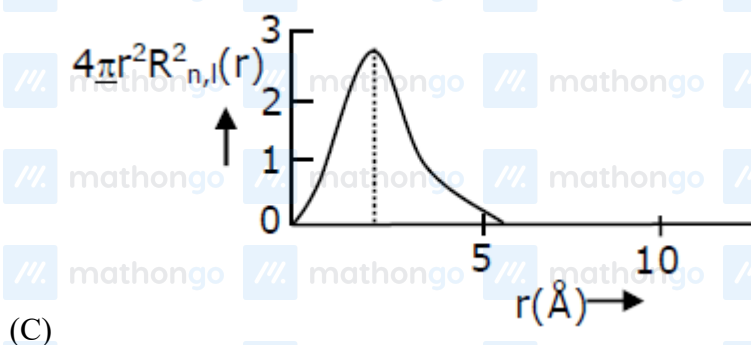
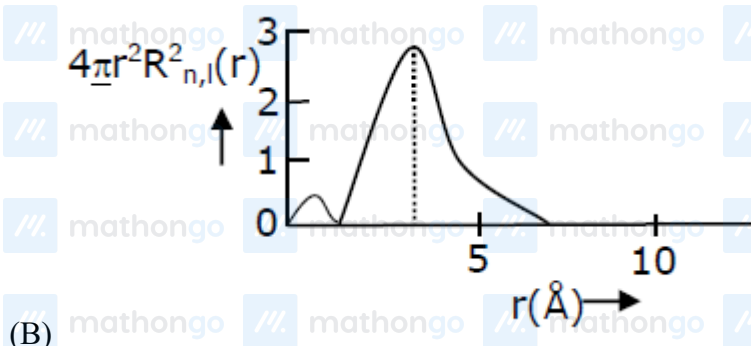
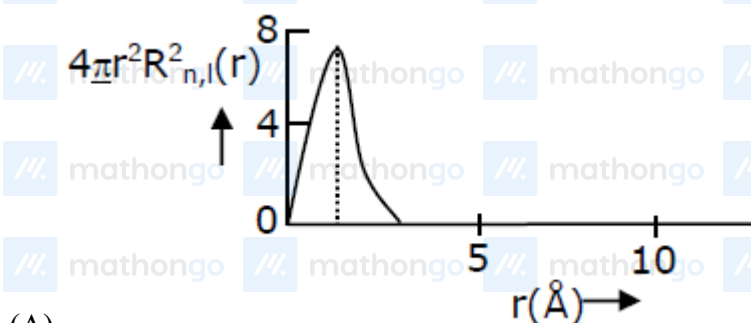
According to Bohr's atomic theory :

- (A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$
- (B) The product of velocity (v) of electron and principal quantum number (n). ' vn ' $\propto Z^2$.
- (C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.
- (D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$. Choose the most appropriate answer from the options given below:

- (1) (C) only
- (2) (A) and (D) only
- (3) (A) only
- (4) (A), (C) and (D) only

Q4: 25 Feb (Shift 1) - Single Correct

The plots of radial distribution functions for various orbitals of hydrogen atom against 'r' are given below:



Questions with Answer Keys

MathonGo

The correct plot for 3 s orbital is:

- (1) D
- (2) B
- (3) A
- (4) C

Q5: 25 Feb (Shift 2) - Numerical

Electromagnetic radiation of wavelength 663nm is just sufficient to ionize the atom of metal A. The ionization energy of metal A in kJmol^{-1} is _____. (Rounded off to the nearest integer)

$$[h = 6.63 \times 10^{-34} \text{ Js}, c = 3.00 \times 10^8 \text{ ms}^{-1}, N_A = 6.02 \times 10^{23} \text{ mol}^{-1}]$$

Q6: 26 Feb (Shift 1) - Single Correct

The orbital having two radial as well as two angular nodes is

- (1) 5 d
- (2) 4f
- (3) 3p
- (4) 4 d

Q7: 26 Feb (Shift 2) - Numerical

A ball weighing 10 g is moving with a velocity of 90 ms^{-1} . If the uncertainty in its velocity is 5%, then the

uncertainty in its position is _____ $\times 10^{-33} \text{ m}$. (Rounded off to the nearest integer) [Given :

$$h = 6.63 \times 10^{-34} \text{ Js}]$$

Answer Key**Q1 (4)****Q2 (2)****Q3 (2)****Q4 (1)****Q5 (180)****Q6 (1)****Q7 (1)**

Q1 (4)

By theory

Q2 (2)

De Brogir Davelength $\lambda = \frac{h}{\sqrt{2mk.E}}$

$$\frac{\lambda_{Li^{+3}}}{\lambda_p} = \sqrt{\frac{m_p \times (e^-v)_p}{m_{Li^{+3}} \times 3e_p V}}$$

$$m_{Li^{+3}} = 8.3m_p$$

$$\frac{\lambda_{Li^{+3}}}{\lambda_p} = \sqrt{\frac{m_p}{3 \times 8.3m_p}} = \sqrt{\frac{1}{25}}$$

$$= \frac{1}{5} = 0.2 = 2 \times 10^{-1}$$

$$x = 2$$

Q3 (2)

$$(A) KE = -TE = 13.6 \times \frac{Z^2}{n^2} eV$$

$$KE \propto \frac{Z^2}{n^2}$$

$$(B) V = 2.188 \times 10^6 \times \frac{Z}{n} \text{ m/sec}$$

$$\text{So, } Vn \propto Z$$

$$(C) \text{ Frequency} = \frac{V}{2\pi r}$$

$$\text{So, } F \propto \frac{Z^2}{n^3}$$

$$\left[\because r \propto \frac{n^2}{Z} \text{ and } V \propto \frac{Z}{n} \right]$$

$$(D) \text{ Force} \propto \frac{Z}{r^2}$$

$$\text{So, } F \propto \frac{Z^3}{n^4}$$

So, only statement (A) is correct

Q4 (1)

3s orbital

$$\text{Number of radial nodes} = n - \ell - 1$$

$$\text{For 3 s orbital } n = 3, \ell = 0$$

$$\text{Number of radial nodes} = 3 - 0 - 1 = 2 \text{ It is correctly represented in graph of option D}$$

Q5 (180)

$$\text{Energy req. to ionize an atom of metal 'A'} = \frac{hc}{\lambda} = \frac{hc}{663nm}$$

$$\text{for 1 mole atoms of 'A' Total energy required} = N_A \times \frac{hc}{\lambda}$$

$$= \frac{6.023 \times 10^{23} \times 6.63 \times 10^{-34} \times 3 \times 10^8}{663 \times 10^{-9}}$$

$$= 6.023 \times 3 \times 10^{23-34+8+7}$$

$$= 18.04 \times 10^4 \text{ J/mol}$$

$$= 180.4 \text{ KJ/mol}$$

$$\text{Nearest Integer} = 180 \text{ KJ/Mol.}$$

Q6 (1)

$$A \cdot N =$$

$$R \cdot N = n - \ell - 1$$

Orbital	Angular Node	Radial Node
5 d	2	2
4 f	3	0
3 p	1	1
4 d	0	1

Q7 (1)

$$m = 10g = 10^{-2} \text{ Kg}$$

$$v = 90 \text{ m/sec}$$

$$\Delta v = v \times 5\% = 90 \times \frac{5}{100} = 4.5 \text{ m/sec}$$

$$m \cdot \Delta v \cdot \Delta x \geq \frac{h}{4\pi}$$

$$10^{-2} \times 4.5 \times \Delta x \geq \frac{6.63 \times 10^{-34}}{4 \times \frac{22}{7}}$$

$$\Delta x \geq \frac{6.63 \times 7 \times 10^{-34}}{9 \times 4 \times 22 \times 10^{-2}}$$

$$\Delta x \geq 1.17 \times 10^{-33} = x \times 10^{-33}$$

$$x = 1.17 \simeq 1$$